Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

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Aim

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR)

values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

Objective – 3: Analyze Antenna Gain (dB)

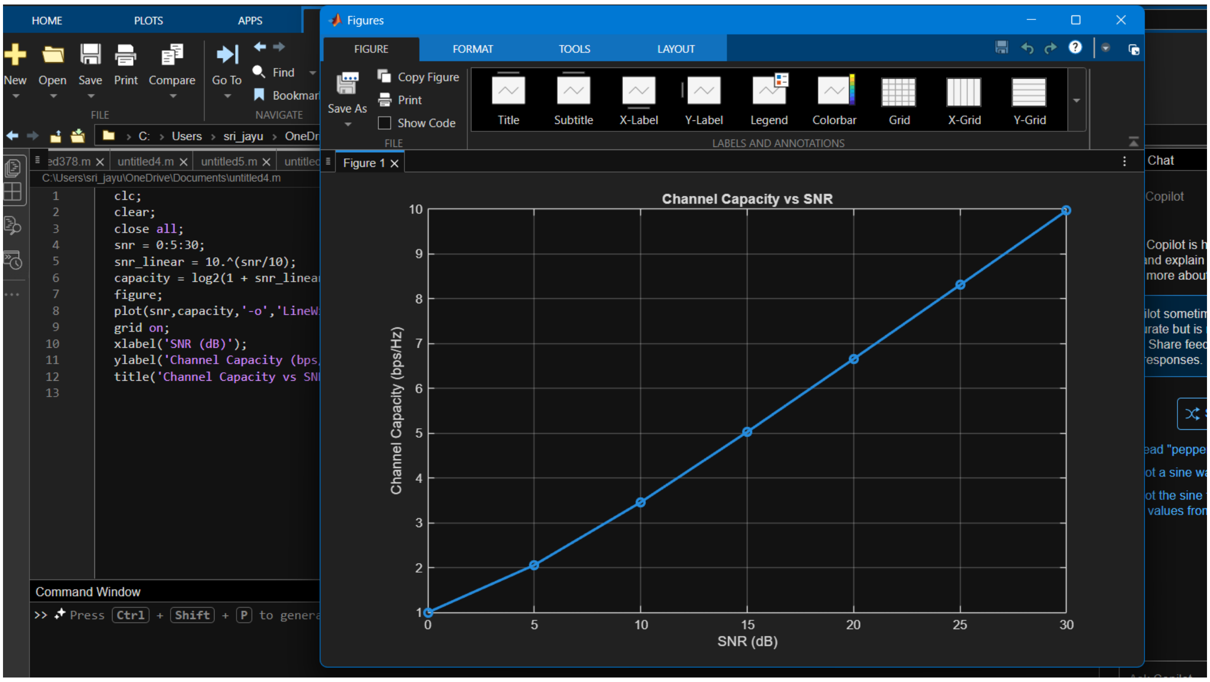
Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

MATLAB Code:

```
clc;

clear;
```

```
close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

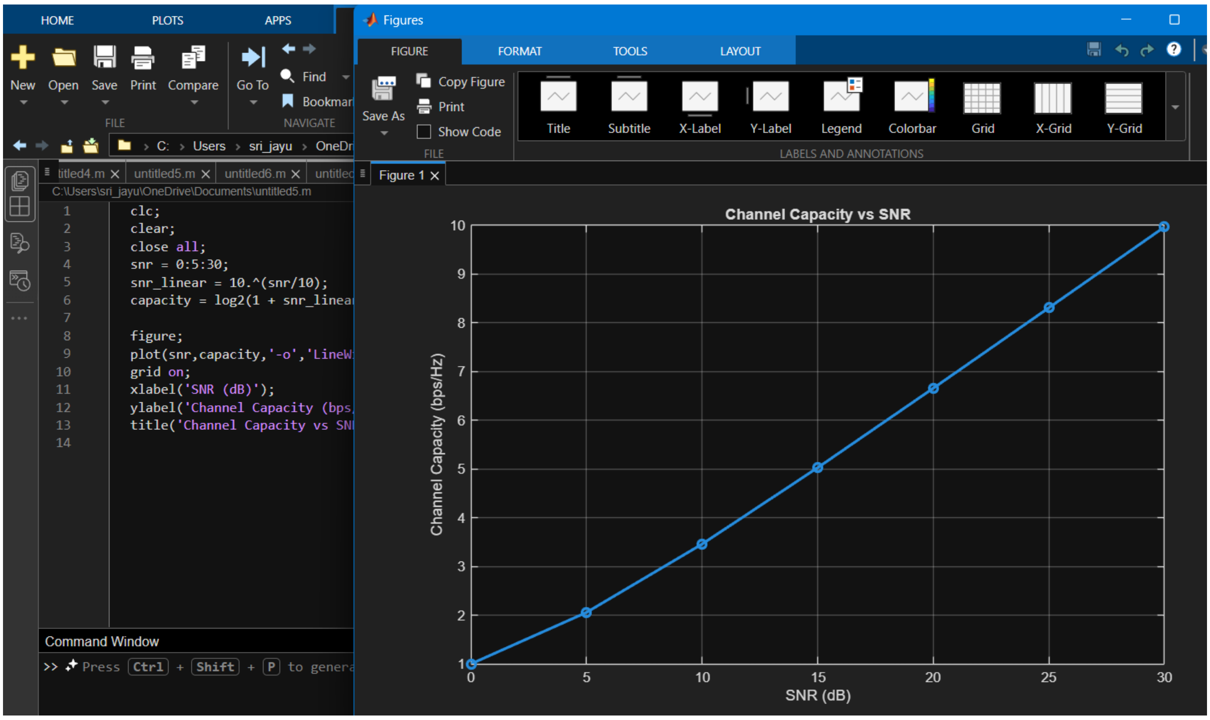
Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR)

values using MATLAB. Observe how channel capacity increases with improving signal quality

based on the Shannon capacity principle.

MATLAB Code:

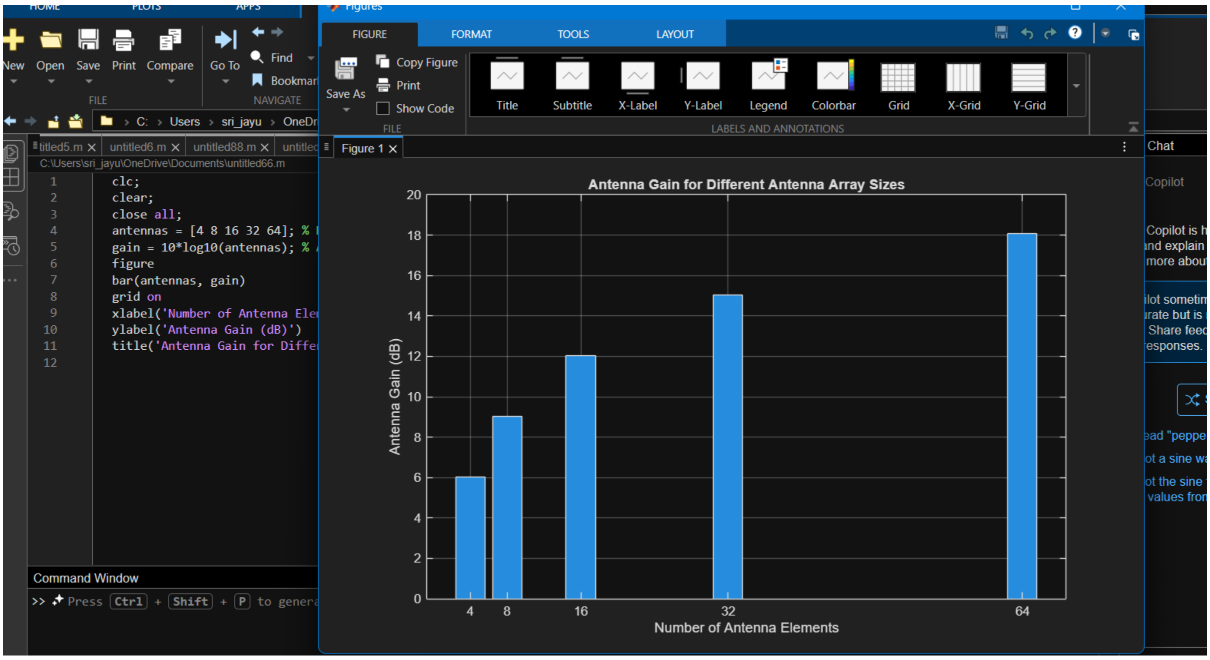
```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

MATLAB Code:

```
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```



Result

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.